

## "Particularity" Injector Waterflow Analysis

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Calculations

| Trial | Pre-Flow ΔP (kPa) | Flow ΔP (kPa) | Correct. ΔP (Pa) | m-Dot (kg/s) | CdA (m²)         | Q-Dot (L/s)  | h Loss (m)  | Eq. Dia (m)    | W-Dot (W)   | Re          | Cavitation |
|-------|-------------------|---------------|------------------|--------------|------------------|--------------|-------------|----------------|-------------|-------------|------------|
| 1     | 0                 | 441.2644668   | 441264.4668      | 0.5785879947 | 0.00001949573761 | 0.5797474897 | 45.08662635 | 0.004982242877 | 255.8219669 | 147603.2189 |            |
| 2     | 0                 | 441.2644668   | 441264.4668      | 0.5734210597 | 0.00001932163582 | 0.5745702001 | 45.08662635 | 0.004959946652 | 253.537413  | 146942.6741 |            |
| 3     | 0                 | 448.1592241   | 448159.2241      | 0.5520010595 | 0.00001845625037 | 0.553107274  | 45.79110489 | 0.004847600212 | 247.8801267 | 144731.9524 |            |
| 4     | 0                 | 448.1592241   | 448159.2241      | 0.5716418694 | 0.00001911294423 | 0.5727874443 | 45.79110489 | 0.004933087918 | 256.6999766 | 147284.3087 |            |
| 5     | 0                 | 441.2644668   | 441264.4668      | 0.5815553551 | 0.00001959572393 | 0.5827207967 | 45.08662635 | 0.004995002564 | 257.1339816 | 147981.2356 |            |
| 6     | 0                 | 461.9487386   | 461948.7386      | 0.5580560673 | 0.00001837810265 | 0.5591744162 | 47.20006196 | 0.004837326437 | 258.3099162 | 146630.307  |            |
| 7     | 0                 | 448.1592241   | 448159.2241      | 0.560209958  | 0.00001873071631 | 0.5613326232 | 45.79110489 | 0.004883511923 | 251.5663929 | 145804.1473 |            |
| 8     | 0                 | 234.421748    | 234421.748       | 0.4293837885 | 0.00001985027031 | 0.430244277  | 23.95227025 | 0.005027340164 | 100.8586155 | 108557.211  |            |
| 9     | 0                 | 289.5798063   | 289579.8063      | 0.4903882112 | 0.00002039743774 | 0.4913709531 | 29.58809854 | 0.005096157802 | 142.2911054 | 122306.1961 |            |
| 10    | 0                 | 275.7902917   | 275790.2917      | 0.4793955288 | 0.00002043262778 | 0.4803562413 | 28.17914147 | 0.005100551901 | 132.4775879 | 119461.5412 |            |
| 11    | 0                 | 248.2112626   | 248211.2626      | 0.4434703645 | 0.0000199238651  | 0.4443590827 | 25.36122732 | 0.005036650964 | 110.2949289 | 111911.3288 |            |
| 12    | 0                 | 289.5798063   | 289579.8063      | 0.4744471916 | 0.00001973437948 | 0.4753979876 | 29.58809854 | 0.005012643249 | 137.6656572 | 120301.8729 |            |
| 13    | 0                 | 151.6846604   | 151684.6604      | 0.3676695863 | 0.00002113034588 | 0.3684063991 | 15.49852781 | 0.005186905818 | 55.88159955 | 90094.99166 |            |
| 14    | 0                 | 110.3161167   | 110316.1167      | 0.3307444365 | 0.00002228911353 | 0.331407251  | 11.27165659 | 0.005327230121 | 36.55956097 | 78911.88096 |            |
| 15    | 0                 | 0             | 0                | #DIV/0!      | #DIV/0!          | #DIV/0!      | 0           | #DIV/0!        | #DIV/0!     | #DIV/0!     |            |
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| 19    | 0                 | 0             | 0                | #DIV/0!      | #DIV/0!          | #DIV/0!      | 0           | #DIV/0!        | #DIV/0!     | #DIV/0!     |            |
| 20    | 0                 | 0             | 0                | #DIV/0!      | #DIV/0!          | #DIV/0!      | 0           | #DIV/0!        | #DIV/0!     | #DIV/0!     |            |
| 21    | 0                 | 0             | 0                | #DIV/0!      | #DIV/0!          | #DIV/0!      | 0           | #DIV/0!        | #DIV/0!     | #DIV/0!     |            |
| 22    | 0                 | 0             | 0                | #DIV/0!      | #DIV/0!          | #DIV/0!      | 0           | #DIV/0!        | #DIV/0!     | #DIV/0!     |            |
| 23    | 0                 | 0             | 0                | #DIV/0!      | #DIV/0!          | #DIV/0!      | 0           | #DIV/0!        | #DIV/0!     | #DIV/0!     |            |
| 24    | 0                 | 0             | 0                | #DIV/0!      | #DIV/0!          | #DIV/0!      | 0           | #DIV/0!        | #DIV/0!     | #DIV/0!     |            |
| 25    | 0                 | 0             | 0                | #DIV/0!      | #DIV/0!          | #DIV/0!      | 0           | #DIV/0!        | #DIV/0!     | #DIV/0!     |            |
| 26    | 0                 | 0             | 0                | #DIV/0!      | #DIV/0!          | #DIV/0!      | 0           | #DIV/0!        | #DIV/0!     | #DIV/0!     |            |
| 27    | 0                 | 0             | 0                | #DIV/0!      | #DIV/0!          | #DIV/0!      | 0           | #DIV/0!        | #DIV/0!     | #DIV/0!     |            |
| 28    | 0                 | 0             | 0                | #DIV/0!      | #DIV/0!          | #DIV/0!      | 0           | #DIV/0!        | #DIV/0!     | #DIV/0!     |            |
| 29    | 0                 | 0             | 0                | #DIV/0!      | #DIV/0!          | #DIV/0!      | 0           | #DIV/0!        | #DIV/0!     | #DIV/0!     |            |
| 30    | 0                 | 0             | 0                | #DIV/0!      | #DIV/0!          | #DIV/0!      | 0           | #DIV/0!        | #DIV/0!     | #DIV/0!     |            |
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| 40    | 0                 | 0             | 0                | #DIV/0!      | #DIV/0!          | #DIV/0!      | 0           | #DIV/0!        | #DIV/0!     | #DIV/0!     |            |
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| 42    | 0                 | 0             | 0                | #DIV/0!      | #DIV/0!          | #DIV/0!      | 0           | #DIV/0!        | #DIV/0!     | #DIV/0!     |            |
| 43    | 0                 | 0             | 0                | #DIV/0!      | #DIV/0!          | #DIV/0!      | 0           | #DIV/0!        | #DIV/0!     | #DIV/0!     |            |
| 44    | 0                 | 0             | 0                | #DIV/0!      | #DIV/0!          | #DIV/0!      | 0           | #DIV/0!        | #DIV/0!     | #DIV/0!     |            |
| 45    | 0                 | 0             | 0                | #DIV/0!      | #DIV/0!          | #DIV/0!      | 0           | #DIV/0!        | #DIV/0!     | #DIV/0!     |            |
| 46    | 0                 | 0             | 0                | #DIV/0!      | #DIV/0!          | #DIV/0!      | 0           | #DIV/0!        | #DIV/0!     | #DIV/0!     |            |
| 47    | 0                 | 0             | 0                | #DIV/0!      | #DIV/0!          | #DIV/0!      | 0           | #DIV/0!        | #DIV/0!     | #DIV/0!     |            |
| 48    | 0                 | 0             | 0                | #DIV/0!      | #DIV/0!          | #DIV/0!      | 0           | #DIV/0!        | #DIV/0!     | #DIV/0!     |            |

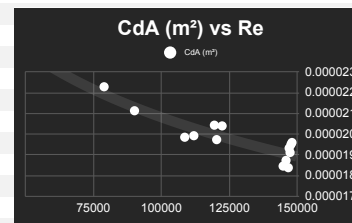
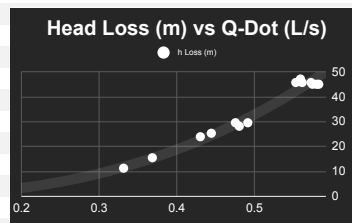
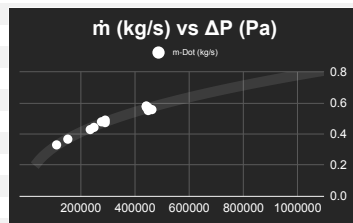
## "Particularity" Injector Waterflow Analysis

## Constants & Notes

[illegible]

## Summary

|                              |                           |                   |                            |                            |                          |                         |                |
|------------------------------|---------------------------|-------------------|----------------------------|----------------------------|--------------------------|-------------------------|----------------|
| <b>Trials</b>                | <b>Mean Cd (m³)</b>       | <b>Cd STDEV</b>   | <b>Cd Variation Coeff.</b> | <b>Eq. Orifice Dia (m)</b> | <b>Mean m-Dot (kg/s)</b> | <b>Mean Q-Dot (L/s)</b> | <b>Mean Re</b> |
| 14                           | 0.00001945247428          | 0.000001057647763 | 3.51%                      | 0.005016157043             | 0.5243798707             | 0.5254307322            | 127,037.35     |
| FULLY TURBULENT              |                           |                   |                            |                            |                          |                         |                |
| <b>Area per orifice (m²)</b> | <b>Area Physical (m²)</b> | <b>L/D Ratio</b>  | <b>Cd</b>                  | <b>Cd Verify</b>           |                          |                         |                |
| 0.0000007853981634           | 0.00002827433388          | 6.35              | 0.6879905414               | OK                         |                          |                         |                |
| 0.6879905415                 |                           |                   |                            |                            |                          |                         |                |



IN NUMERIS VERITAS